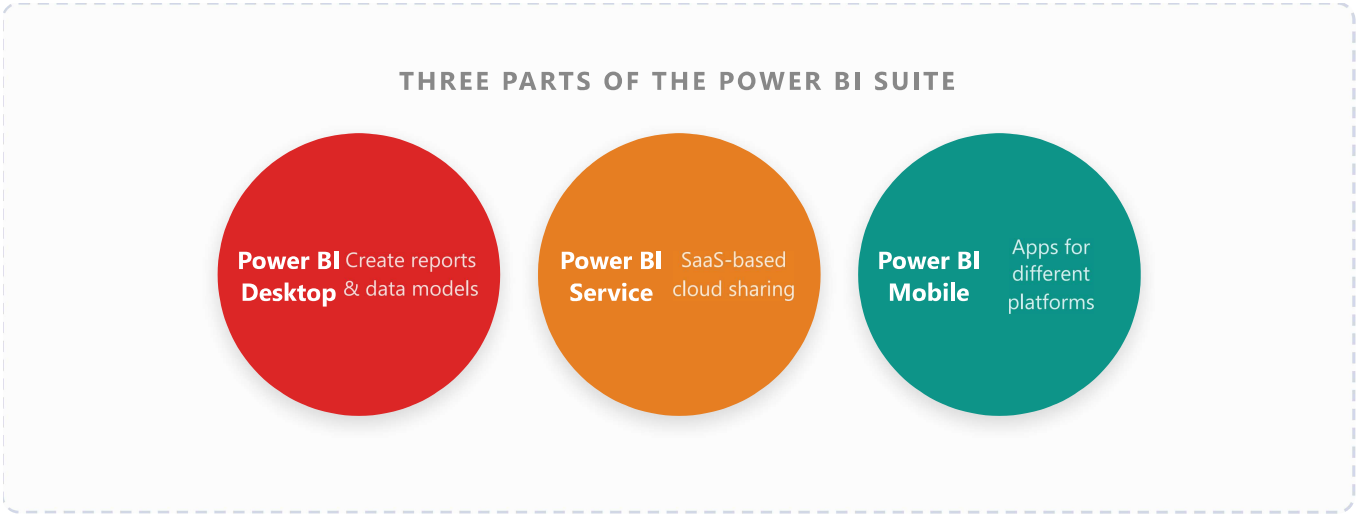
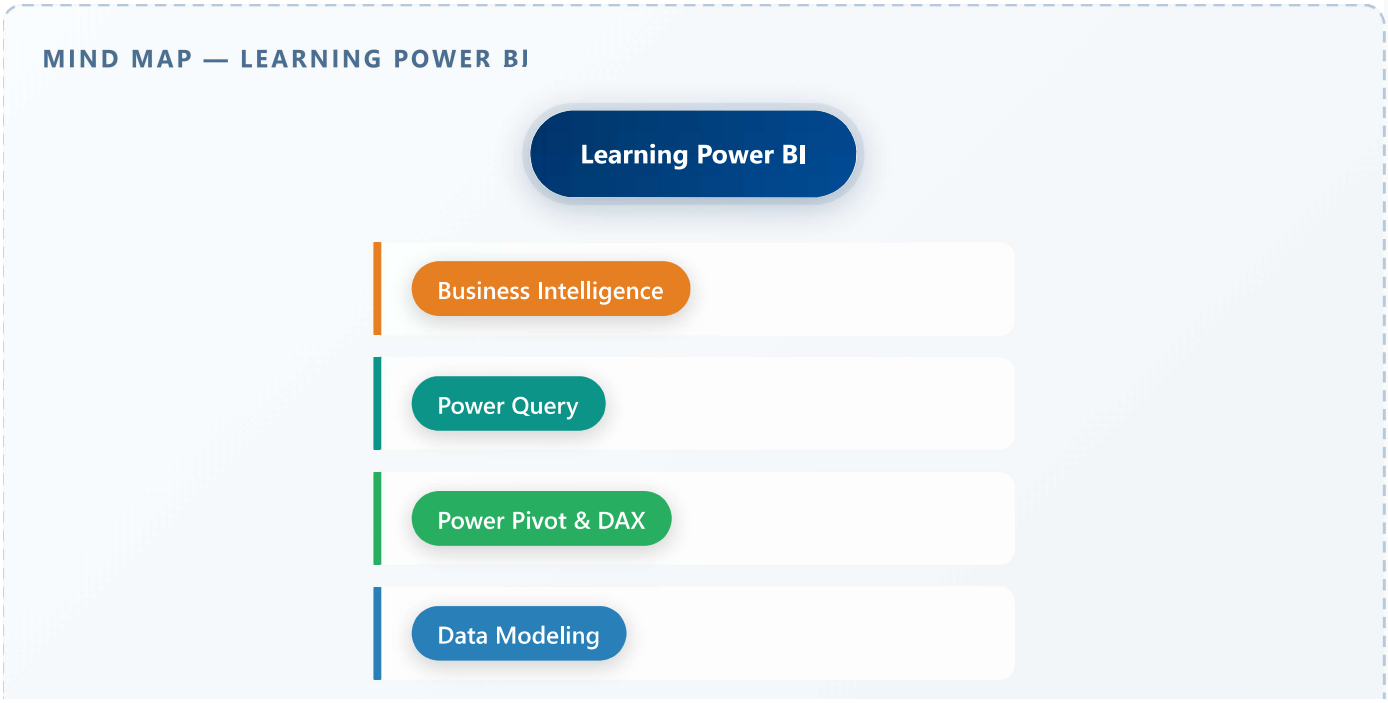
Learning Power BI

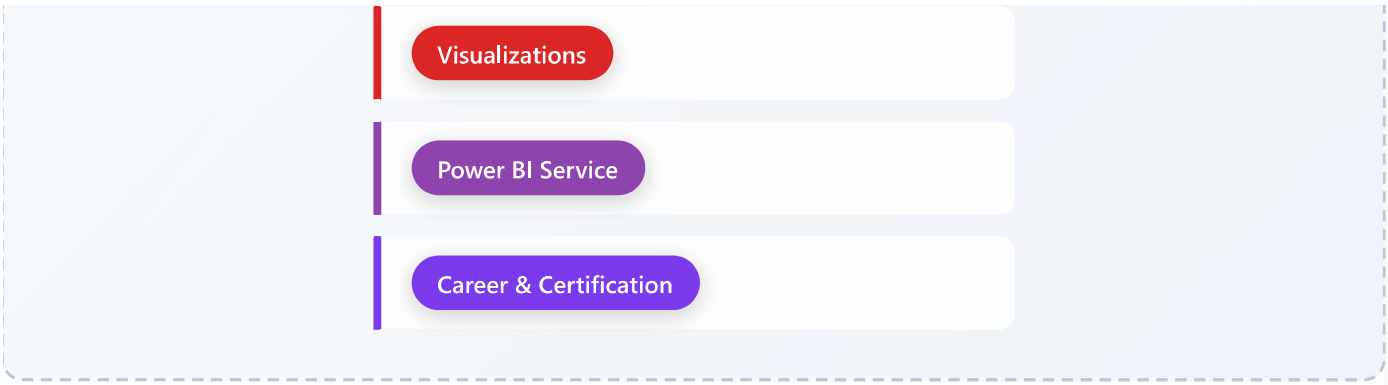
Data Visualization & Business Intelligence — Comprehensive Tutorial

Topic Overview: Learning Power BI

Power BI is a Data Visualization and Business Intelligence tool that converts data from different data sources to interactive dashboards and BI reports. Power BI is a business intelligence tool developed by Microsoft that transforms raw data into interactive dashboards and meaningful insights. Power BI suite provides multiple software, connector, and services — Power BI Desktop, Power BI Service based on SaaS, and mobile Power BI apps available for different platforms. These set of services are used by business users to consume data and build BI reports.







What is Business Intelligence?

Business Intelligence assists companies in processing and filtering massive amounts of stored data and transforming it into actionable insights. Mining, Data Warehouse, and Data Analytics are the correlated names of Business Intelligence. Exploratory data analysis, dashboards, and predictive modeling are the core methods of business intelligence.

Why Do We Need a Business Intelligence Tool?

In today's digital era, unstructured data is dispersed all around the world. Companies face difficulty in handling and monitoring huge amounts of data, which in turn negatively affects their business strategies and operation management. This is why the companies require powerful BI tools for quick decision-making to increase their sales production, increase the availability of structured data, generate immense reports, handle customers, track employee performance, and enhance operational efficiency.

What Power BI Allows Users to Do

- **Connect to multiple data sources** — import data from Excel, SQL, CSV, JSON, APIs and websites
- **Clean and transform data** — without heavy coding, using the Query Editor
- **Build data models** — create relationships between tables for efficient reporting
- **Create powerful visual reports** — interactive charts, KPIs, dashboards and reports
- **Analyze trends** — using filters and slicers
- **Share insights securely** — share dashboards securely with teams
- **Automate data refresh** — schedule automatic data refresh

5
Major Components

3
Suite Parts

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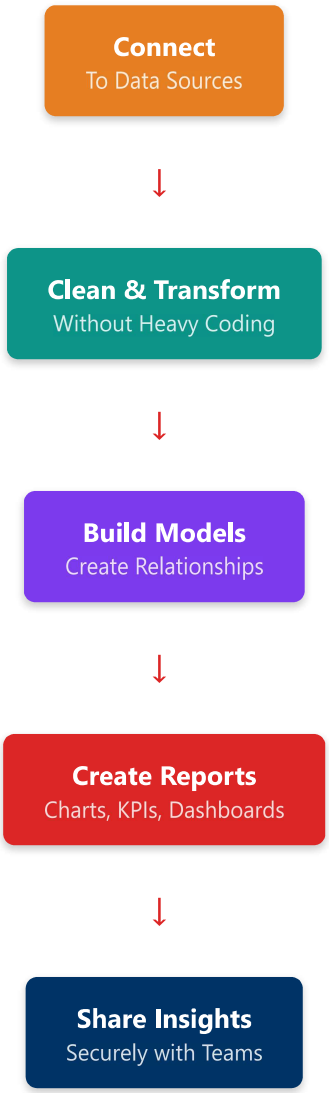
Certification Exam

Many

Domains Using Power BI

Industry Usage

Popular business intelligence tools such as Power BI and Tableau are being used by financial companies and big firms like Deloitte, PwC, and EY to compile data from various sources and present it in an attractive format that can be used for making crucial business decisions and identifying market trends and patterns.



Prerequisites to Learn Power BI

Before proceeding with this tutorial, you should be familiar with Microsoft Excel, data modeling, and have some knowledge of DAX language. Anyone who is well versed in Excel can easily learn Power BI.

Major Components of Power BI

Power BI has five major components, each serving a distinct purpose in the data-to-insight pipeline. Understanding these components is foundational to learning Power BI.

Interactive Dashboard: Five Major Components

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal each component's details.

Power BI Major Components Dashboard 5 / 5 Placed

Power Query

- Process of cleansing and transforming data
- Permits users to access datasets connecting from multiple sources
- Included on the Power BI Desktop
- Business users may view data from MySQL, SQL servers, DB2, and many more

Power View

- A data visualization tool
- Assists users in developing stunning charts and colorful maps
- Turns data into a story
- Helps create visually appealing reports

Power Map

- A 3D map visualization tool
- Identifies geospatial data on Map visuals
- Examines maximum sales production geographically
- Visualizes the demographic populations of specific regions

Power Pivot

- A Data Modelling technique
- Used to create relationships between datasets
- Performs complex computations by utilizing DAX functions
- Foundation for building data models in Power BI

Power Q & A

- When dealing with giant datasets, crucial for in-depth details
- Done through natural language
- Users may ask questions and obtain the answer through Power Q&A
- Enables conversational data exploration

Power BI Architecture

Power BI architecture provides an overview of the Power BI components such as Power BI Desktop, Gateway, Power BI Mobile Apps, and Power BI Service. These components work together to transform raw data into interactive dashboards and meaningful insights.

Interactive Dashboard: Architecture Components

Click a component from the pool below, then click the matching gray zone to reveal each architecture element.

Power BI Architecture Dashboard 5 / 5 Placed

Data Sources

- Excel, SQL, CSV, JSON, APIs, websites
- Flat files and SQL databases
- Azure Cloud platform data sources
- Online services like Salesforce, SharePoint

Power BI Desktop

- Three views: Report View, Data View, and Model View
- Power Query Editor for data cleaning and transformation
- Power Pivot for data modelling and DAX
- Where reports and data models are built

Data Gateway

- Bridge between on-premises data and Power BI Service
- Enables scheduled data refresh
- Manages data source settings
- Connects on-premises data sources to the cloud

Power BI Service

- Cloud-based SaaS service for publishing and sharing
- Workspaces for organizing content
- Row Level Security (RLS) for data access control
- App publishing and Self-Service BI (SSBI)

Power BI Mobile

- Mobile apps available for different platforms
- View and interact with reports and dashboards on the go
- Touch-optimized interfaces
- Ensures insights are accessible anywhere

AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.

Dashboard Complete!

You've successfully mapped all five Power BI architecture components.

Reset Dashboard

Power BI Desktop: Three Views

The Power BI Desktop has three views that serve different purposes in the report-building process:

The Report View is where you build your reports. It contains the Visualizations pane for selecting chart types, the Data pane for dragging fields, the Filters pane for applying filters, and the canvas area where visuals are placed and arranged on report pages.

The Data View lets you see the data in your tables in a tabular format, similar to Excel. Here you can inspect data, create calculated columns, and manage data types for individual columns before building visuals.

The Model View shows a diagram of all your tables and the relationships between them. This is where you create, edit, and manage table relationships — connecting fact tables to dimension tables to build a data model.

Power BI Pricing Versions

The tutorial covers three pricing versions of Power BI:

- **Power BI Free:** For individual use with basic features; allows report creation in Desktop but limited sharing capabilities
- **Power BI Pro:** For teams and small organizations; enables sharing dashboards securely with colleagues, collaboration, and more data refresh options
- **Power BI Premium:** For large organizations with advanced needs; provides increased capacity, larger data storage, and enhanced features at the enterprise level

Power BI Workflow Steps

Power BI allows users to follow a structured workflow from connecting to data sources through to sharing insights securely with teams. The following five-step workflow represents the core process covered in the Power BI tutorial.



Power BI supports connecting to multiple data sources. As covered in the tutorial, supported data sources include flat files (Excel, CSV, JSON), SQL databases, the Azure Cloud platform, and online services. Users can import data from Excel, SQL, CSV, JSON, APIs and websites. This is the first step in the Power BI workflow — identifying and connecting to the relevant data sources for your analysis.

Using the Power Query Editor, users can clean and transform data without heavy coding. The tutorial covers Data Types, Working with Numbers, Working with Date and Time, creating Conditional Columns, Grouping and Aggregating Records, Merge and Append Queries, managing data source settings, Data Refresh, and Excel Integration. The concept of data extraction and transformation in the Power Query Editor is also covered, including identifying data discrepancies.

Data Modeling involves understanding Data Tables vs. Lookup Tables, Creating Table Relationships, Editing Table Relationships, Managing Active vs. Inactive Relationships, Connecting Multiple Data Tables, Understanding Filters, and Hiding tables, columns and fields from Power Pivot. The tutorial also covers how to develop a Star Schema — how a fact table is connected to multiple dimensional tables.

This step covers creating dashboards and reports using charts, slicers and filters. The tutorial covers Creating Dashboard, Creating Reports, Basic Charts and Visuals, Conditional Formatting, adding Reports to Dashboards, adding Trend Lines and Forecasts, Report Filtering Options, exploring Data with Matrix Visuals, Filtering with Date Slicers, Editing Power BI Report Interactions, and Adding Drillthrough Filters.

Power BI Service allows you to publish, share and manage reports in the cloud. The tutorial covers Publishing Reports, Workspaces, Row Level Security (RLS), Data Gateway, How to Publish App, and Self-Service Business Intelligence (SSBI). Sharing dashboards with other users and exploring datasets are also covered.

Power BI Desktop vs Service

The tutorial covers the differences between the Power BI Desktop and the Power BI Service and their different phases. The Desktop is where you build reports and data models, while the Service is where you publish, share, and manage those reports in the cloud. Dashboard options such as "Get Quick Insights," "Pin to Dashboard," and "Tiles in Dashboard" are features of the Power BI Service, not the Desktop.

DAX & Data Modeling

DAX (Data Analysis Expressions) is introduced for creating measures and calculations in Power BI. The tutorial covers row context and filter context — two fundamental concepts for understanding how DAX evaluates expressions — along with commonly used DAX function categories.

DAX Function Categories Covered in the Tutorial

DAX aggregate functions perform calculations across a set of values and return a single value. The tutorial covers the overview of the DAX function and the different categories of DAX aggregate functions, along with their examples. These are the most commonly used functions for creating measures that sum, average, count, or otherwise aggregate data values.

The tutorial explores multiple types of DAX Date functions, such as the HOUR, WEEKDAY, WEEKNUM, and NOW functions. Date and Time functions are essential for time-based analysis and are used extensively in creating time-intelligence calculations for reports.

The tutorial details the numerous types of DAX Logical functions along with their examples. Logical functions evaluate conditions and return different results based on whether the condition is true or false, enabling conditional logic within DAX calculations.

- **Text & Window Functions:** For manipulating text strings and performing window-based calculations
- **Information Functions:** Such as CONTAINS, CONTAINSSTRING, CONTAINSSTRINGEXACT — used to check values and metadata
- **Math Functions:** For mathematical operations and calculations
- **Index Functions:** For indexing and referencing specific rows in a table

- **COUNT Functions:** For counting rows, distinct values, and other counting operations — types of DAX counting functions are covered with examples
- **Depreciation Functions:** Detailed study of the Depreciation Functions and their types — used for financial calculations

Row Context and Filter Context

Core DAX Concepts

Two fundamental concepts in DAX covered in the tutorial are **Row Context** and **Filter Context**. Row Context refers to the environment where DAX evaluates each row individually — this is the context used when creating calculated columns. Filter Context refers to the environment determined by filters applied through slicers, rows, columns, and report/page/visual-level filters — this is the context used when evaluating measures. Understanding the difference between these two contexts is essential for writing correct DAX calculations.

Data Modeling Concepts

Star Schema and Relationships

The tutorial covers how to develop a Star Schema — how a fact table is connected to multiple dimensional tables. Data modeling also includes understanding the difference between Data Tables and Lookup Tables, creating and editing table relationships, managing Active vs. Inactive Relationships, connecting multiple data tables, understanding filters, and hiding tables, columns and fields from Power Pivot. Multiple types of cardinalities in Power BI Desktop are also discussed.

Important Prerequisite

Before proceeding with DAX and Data Modeling, you should have some knowledge of DAX language and be familiar with data modeling concepts. The tutorial is prepared for beginners to help them understand the basic concepts, from where you can take yourself to a higher level of expertise.

Visualization Types in Power BI

Power BI provides a wide range of visualization options. The tutorial explores various visualization types including Simple visualization, Map visualization, and combination charts. The following are the specific visualization chart types covered in the Power BI tutorial.

Complete List of Visualization Charts Covered

- **Stacked Bar Chart** — learn how to create an interactive Stacked bar chart
- **Stacked Column Chart** — basic concept and creation in the BI desktop
- **Clustered Chart** — master proficiency of generating the Clustered chart
- **100% Stacked Chart** — deep insights into the 100% Stacked Chart
- **Area Chart and Stacked Area Chart** — overview and creation in the BI desktop
- **Line and Stacked Column Chart** — fundamental concept with step by step explanation
- **Line and Clustered Column Chart** — creating interactive Line and Clustered Column Charts
- **Ribbon Chart** — basic concept and creation for interactive reports
- **Table and Matrix Visuals** — fundamental concept of generating Table and Matrix visuals

Map Visualizations

The tutorial covers multiple types of Map visualizations and explains how to generate the MAP, Filled Map, and TreeMap visualizations in the BI report. Additionally, the ArcGIS Map is covered — learn how to develop the ArcGIS Map by signing up for ArcGIS Online or adding the location-based field to the Location section. Power Map is described as a 3D map visualization tool to identify geospatial data on Map visuals, seamlessly helping organizations examine maximum sales production geographically and visualize demographic populations of specific regions.

Specialized Visualizations

- **Waterfall Charts:** Covers the benefits and steps to developing the interactive Waterfall chart by placing crucial fields into the perspective sections in the Data Visualization pane

- **Funnel Charts and Radial Gauge Chart:** Deep insights into generating Funnel charts and Radial Gauge Charts on the BI desktop
- **Scatter Chart:** Fundamental concept and how it is created in the Power BI report

- **Pie Chart and Donut Chart:** How to generate Pie Chart and Donut chart in the BI report
- **Card and Slicer Visualization:** Evaluating values using Card Visualization and filtering data according to categories using Slicers
- **KPI Visual:** How to create the KPI visual
- **Smart Narrative Visual:** In-depth study and how to utilize it along with other visuals effectively
- **Decomposition Tree:** Understanding the Decomposition Tree and its construction
- **Multi Row Card:** Basic overview and detailed study of constructing the multi row card

Additional Features

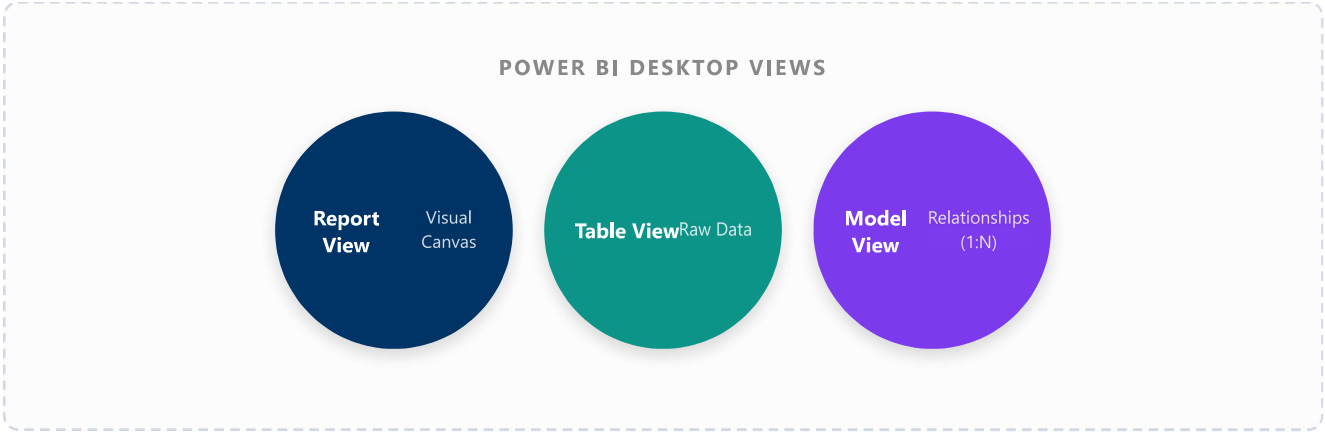
- **Paginated Report:** Basic overview and steps to generate a paginated report in the report builder tool and the Power BI desktop
- **Python Script & R Script:** How to use Python Script and R script on the Power BI desktop
- **Power Apps and Power Automate:** Overview of Power Apps and Power Automate integration with Power BI
- **Excel Integration:** How to import the Excel dataset into the Power BI service
- **Dashboard Options:** "Get Quick Insights," "Pin to Dashboard," "Tiles in Dashboard," and other dashboard options

Business Intelligence — Power BI Desktop

Financial Data Transformation & ETL — Power BI Desktop for Accounting & Finance

Unit 1: Overview of Power BI Desktop Interface

Power BI Desktop is a business intelligence and analytics application used by corporate finance teams, auditors, and financial managers to convert raw financial data into interactive dashboards.



Key Workspaces (Left Navigation Bar)

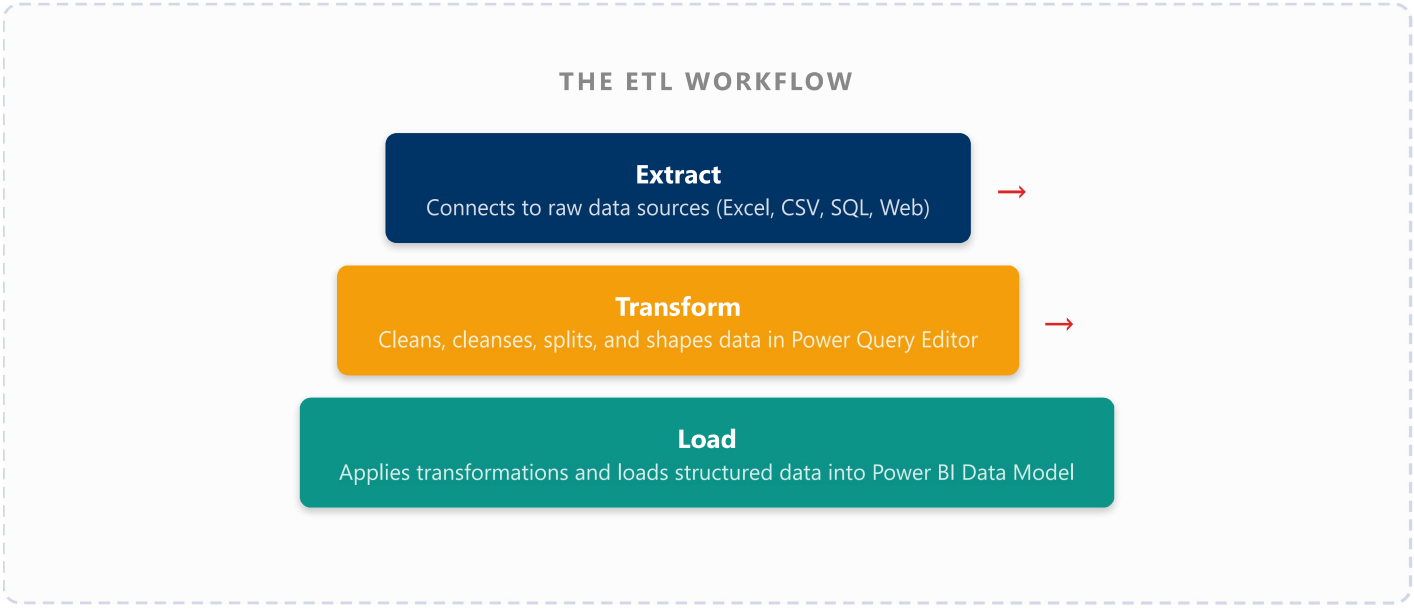
1. **Report View:** The visual canvas where finance dashboards, KPI cards, bar charts, and slicers are designed.
2. **Table View (Data View):** Displays underlying tabular data, row-level records, DAX calculated columns, and measures.
3. **Model View:** Displays entity-relationship diagrams (ERDs) connecting multiple tables (e.g., linking a `General Ledger` table to a `Chart of Accounts` dimension table via Primary Key / Foreign Key relationships).

Key Workspace Panes (Right Navigation Bar)

- **Fields / Data Pane:** Lists imported data tables and their respective columns/measures.
- **Visualizations Pane:** Contains visual types (Stacked Bar, Line Chart, Matrix, KPI Card) and formatting options.
- **Filters Pane:** Applies global, page-level, or visual-level filtering criteria.

Importing Data into Power BI Using Power Query

Raw accounting data usually lives in disparate formats—such as Excel spreadsheets, CSV sales exports, or cloud SQL databases. **Power Query** serves as the **ETL (Extract, Transform, Load)** engine built into Power BI Desktop.



Interactive Dashboard: ETL Workflow

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal each step's details.

Power Query ETL Workflow 3 / 3 Placed

Extract

- Navigate to Home Tab
- Click Get Data
- Select source type (Excel, CSV, SQL)
- Browse and select file (e.g., Monthly_Trial_Balance_2026.xlsx)
- Preview the table or sheet in Navigator Dialog Window

Transform

- Critical Decision Point: Choose Transform Data over Load
- Opens the Power Query Editor to clean and restructure data prior to modeling
- Allows cleaning of metadata rows and system titles

Load

- Directly pushes raw, uncleaned data into the model (not recommended)
- Applies transformations and loads structured data into Power BI Data Model

Fixing Column Names and Header Alignment

Accounting data exported from legacy ERP systems (e.g., Tally, SAP, or legacy accounting packages) often includes metadata rows, system titles, or blank header rows that mess up standard data structures.

RAW EXPORT (MESSY)

Row 1	`ABC Corp - Ledger`
Row 2	`Printed: 17/09/26`
Row 3	`Code | Name | Debit`

↓

POWER QUERY CLEANING

Column 1	`Account Code`
Column 2	`Account Name`
Column 3	`Debit Amount`

1. Demoting / Promoting Headers

When importing raw ledger extracts, the actual column headers (`Account Code`, `Account Name`, `Debit`, `Credit`) often appear down on Row 3 or 4, while Power BI assigns generic titles like `Column1`, `Column2`.

Fix Execution:

1. Navigate to **Home Tab** (in Power Query Editor).
2. If header rows are buried beneath system titles: Use **Remove Top Rows** (specify the number of top operational title rows to delete, e.g., `2`).
3. Click **Use First Row as Headers** (or Home Tab $\rightarrow$ *Transform* group $\rightarrow$ *Use First Row as Headers*).

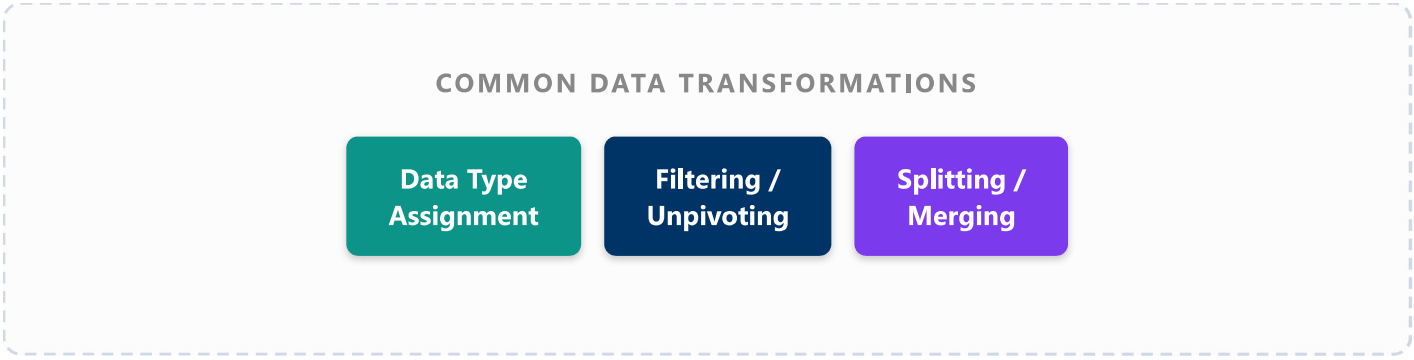
2. Renaming Columns for Financial Standards

Double-click any column header (or right-click $\rightarrow$ **Rename**). Rename unclear ERP shorthand into standard accounting terminology:

- Change `GL_ACC_ID` $\rightarrow$ `Account Code`
- Change `AMT_INR` $\rightarrow$ `Transaction Amount (INR)`

Unit 4: Data Transformation in Power Query

Financial data must be structured as normalized tables (flat rows and columns) before building reports.



1. Assigning Data Types

🚦 **Incorrect data types cause calculation errors.**

- **Text (ABC):**
  Assigned to codes, account IDs, voucher numbers (e.g., `Voucher No: 10045`).
- **Decimal Number (1.2) / Fixed Decimal:**
  Assigned to financial values.
- **Date:**
  Assigned to posting dates (`Transaction Date`).

2. Unpivoting Columns (Converting Wide Data to Tall Data)

Financial balance sheets or budgets are frequently formatted horizontally across years or months (Wide Format), which hinders automated analytics.

Problem Scenario: A budget sheet has columns: `Account`, `Jan_Budget`, `Feb_Budget`, `Mar_Budget`.

Transformation:

1. Select the `Account` column.
2. Right-click $\rightarrow$ Select **Unpivot Other Columns**.
3. Power Query restructures the table into three tall columns: `Account`, `Attribute` (Month Name), and `Value` (Budget Amount).

3. Handling Null Values & Replacing Values

Right-click a column $\rightarrow$ **Replace Values**. Replace `null` entries in expense columns with `0` to prevent mathematical calculations from breaking during aggregation.

4. Splitting Columns & Text Transformations

Accounting Scenario: An ERP exports combined strings like `1010-Cash In Hand`.

Fix: Select Column $\rightarrow$ Click **Split Column** (Home Tab) $\rightarrow$ **By Delimiter** $\rightarrow$ Choose - (Dash). Output: Splits cleanly into `Column 1: Account Code` (`1010`) and `Column 2: Account Name` (`Cash In Hand`).

5. Applied Steps Pane (Audit Trail)

On the right-hand side of the Power Query window, the **Applied Steps** pane logs every transformation chronologically (e.g., *Source*, *Promoted Headers*, *Changed Type*, *Unpivoted Columns*).

- Acts as an **automated audit trail**.
- Clicking the x next to any step deletes that specific action without ruining the raw source file.

Summary Matrix of Power Query Transformations

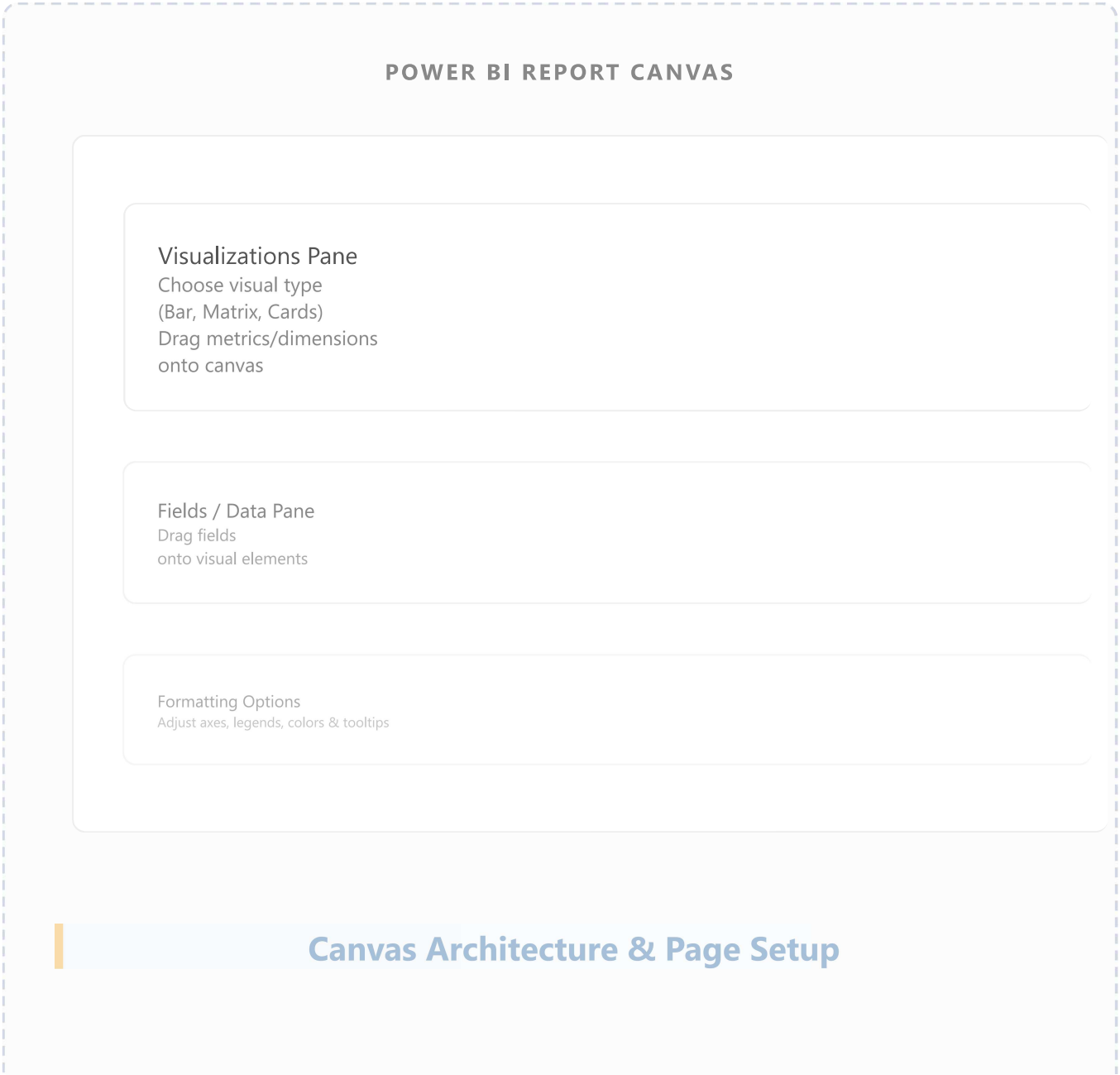
#	Power Query Command	Primary Location	Accounting & Audit Application
1	**Get Data**	Home Tab (Main Window)	Connects Power BI to Excel files, CSVs, or ERP databases
2	**Use First Row as Headers**	Home / Transform Tab	Promotes ledger field names into column headers
3	**Remove Top Rows**	Home Tab $\rightarrow$ Reduce Rows	Clears non-standard ERP export titles & metadata
4	**Change Data Type**	Column Header Icon	Sets numbers to Fixed Decimal (`₹`) and IDs to Text
5	**Unpivot Columns**	Transform Tab	Converts monthly budgeted matrix tables into vertical transactional rows
6	**Split Column**	Transform Tab	Separates combined account strings (e.g., `Code-Name`)
7	**Applied Steps**	Right-side Pane	Reviews and edits the automated data transformation audit trail

Financial Data Visualization — Power BI Desktop

Business Analytics & Data Visualization — Step-by-Step Report Building Guide

Unit 1: Creating Reports in Power BI Desktop

In modern corporate finance and auditing, management teams rely on visual reporting systems rather than static printed ledgers. A **Power BI Report** is a multi-perspective visual display of a structured dataset, consisting of dynamic charts, tables, cards, and slicers laid out on a multi-page canvas.



Canvas Architecture & Page Setup

1. **Canvas Grid:** The central workspace where visual elements are arranged. Standard canvas size defaults to a `16:9` widescreen layout.

2. **Report Pages:** Located at the bottom tab bar. Structured by domain:
   - *Page 1:* Executive Financial Summary (High-level KPIs)
   - *Page 2:* Revenue & Profitability Analysis
   - *Page 3:* Working Capital & Cash Flow Liquidity
   - *Page 4:* Accounts Receivable Aging & Expense Breakdown.

Primary Financial Visuals

Below is a visual reference for the standard visuals used in accounting and finance.

VISUAL REFERENCE

Visual Type: Stacked / Clustered Bar Chart

Use: Comparing discrete financial categories

Example: Total operating expenses across various company departments

Visual Type: Matrix Visual

Use: Cross-tabulating structured financial tables

Example: Income Statement format (Rows: Category, Columns: Quarter, Values: Amount)

Visual Type: Waterfall Chart

Use: Analyzing sequential gains and losses

Example: Bridge chart showing Net Profit derivation from Gross Revenue minus Costs

Visual Type: Donut / Pie Chart

Use: Displaying proportional asset or expense allocations

Example: Portfolio asset allocation percentage or budget expenditure breakdown

Creating Reports in Power BI Desktop

Step-by-Step Practical Exercise: Building a Financial Performance Report

Business Scenario: You are provided with a clean dataset containing `Sales Transactions`, `Operating Costs`, `Target Budget`, and `Branch Offices`.

Build Your Report 3 / 6 Phases Completed



Insert KPI Cards

Drag a Card Visual onto the top left of the canvas

Assign 'Total Sales Revenue'

Drag a KPI Visual onto the canvas

Set Value = 'Actual Profit'

Set Target Goal = 'Budgeted Profit'

Set Trend Axis = 'Month'



Build Trend Visuals

Insert a Line and Stacked Column Chart

Set Shared Axis = 'Fiscal Month'

Column Values = 'Sales Revenue'

Line Values = 'Net Profit Margin %'



Interactive Controls

Add a Slicer for Branch Office

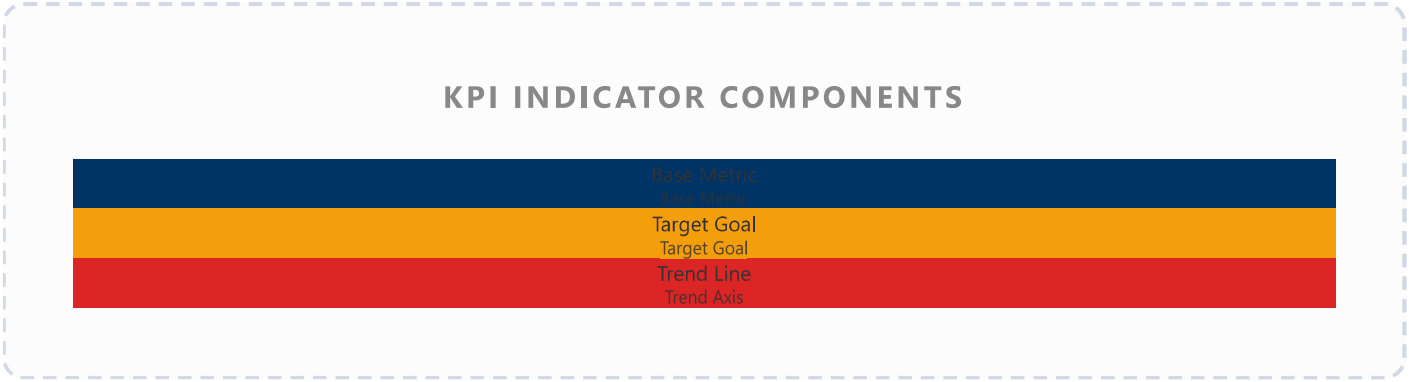
Add a Date Slicer set to *Between* mode

Format it as a horizontal tile row across the top navigation bar

Allows users to filter financial views by Branch, Fiscal Year, or Quarter

Power BI Key Performance Indicators (KPIs) & Card Visuals

A **Key Performance Indicator (KPI)** is a visual cue that communicates the amount of progress made toward a defined quantifiable financial target.



Components of a Power BI KPI Visual

- **Indicator (Base Value):** The current actual performance metric (e.g., `Actual Gross Revenue = ₹ 45,00,000`).
- **Target (Goal):** The planned budget or forecast benchmark (e.g., `Budgeted Target = ₹ 50,00,000`).
- **Trend Axis:** A chronological sequence (e.g., `Fiscal Month`) that plots background trendlines to show performance direction over time.

Benefits of KPI Visuals for Financial Decision-Making

- **Instant Performance Feedback:** Color-coded status displays immediately inform managers whether a financial metric is on track (Green) or off target (Red).
- **Variance Tracking:** Automatically calculates and displays absolute variance (`₹ -5,00,000`) and percentage variance (`-10%`) against budgeted targets.
- **Executive Summary Efficiency:** Replaces large grid tables with scannable executive cards for board presentations and audit reviews.

Building a Financial Performance Report

Use the interactive tool below to assemble the financial performance report described below.

Interactive Report Builder

0 / 3 Phases Completed



Insert KPI Cards

Drag a Card Visual onto the top left of the canvas

Assign `Total Sales Revenue`

Drag a KPI Visual

Set Value = `Actual Profit`

Set Target Goal = `Budgeted Profit`

Set Trend Axis = `Month`



Build Trend & Distribution Visuals

Insert a Line and Stacked Column Chart

Set Shared Axis = `Fiscal Month`

Column Values = `Sales Revenue`

Line Values = `Net Profit Margin %`



Interactive Controls

Add a Slicer for `Branch Office`

Format it as a horizontal tile row

Add a Date Slicer set to *Between* mode

Allows users to filter financial views by Branch, Fiscal Year, or Quarter

Summary Matrix of Report & Visualization Concepts

Quick reference table for accounting workflows.

Cross-Filtering

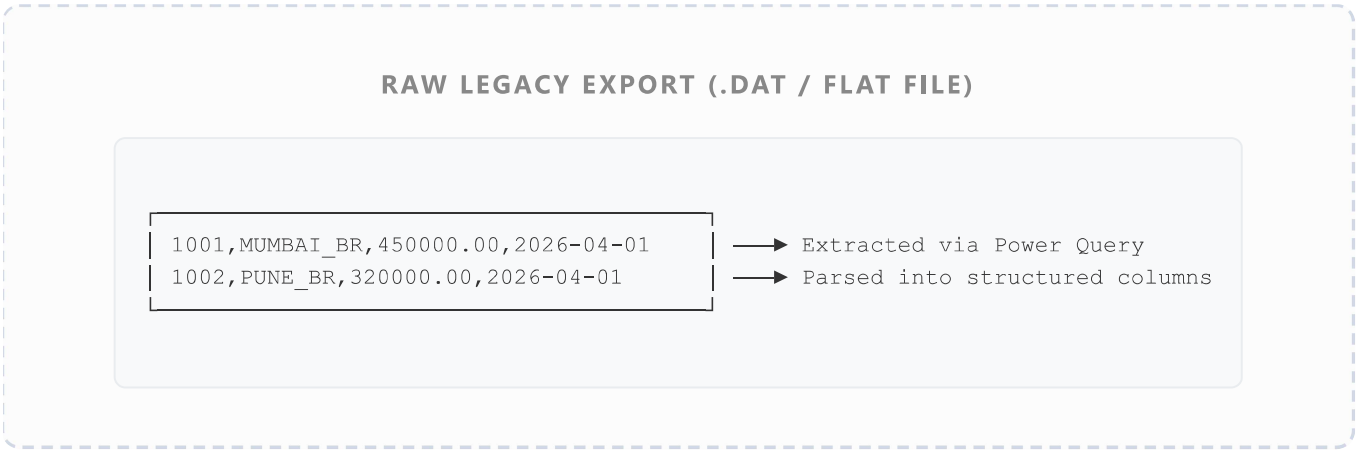
#	Power BI Feature	Visual Element	Purpose in Accounting & Finance
1	**Card Visual**	Single numeric display	Displays key summary figures (e.g., Total Receivables, Total Liabilities)
2	**KPI Indicator**	Card with target & status color	Evaluates actual performance against budgeted targets (Green/Red)
3	**Matrix Visual**	Cross-tabulated grid	Generates structured Balance Sheet and Income Statement layouts
4	**Slicer**	Interactive filter button	Allows users to filter financial views by Branch, Fiscal Year, or Quarter
5	**Waterfall Chart**	Sequential step chart	Displays revenue bridges, gross margin derivation, and cost impact
6	**Inter-chart reactivity**	Automatically updates all visuals on a page when a chart segment is clicked	

Business Intelligence & Cloud Integration

Power BI Desktop & Power BI Service — Second-Year B.Com (Accounts & Finance)

Unit 1: Report Design with Legacy Files & `.DAT` Data Sources

In banking, insurance, and accounting environments, legacy ERP platforms and mainframes often generate flat data exports with extensions like `.DAT` (Data File) or fixed-width text files. These files hold raw transactional records, payroll summaries, or historical general ledger dumps.

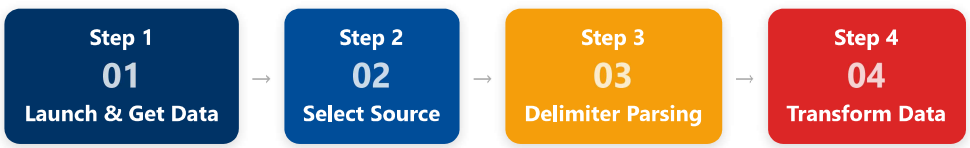


1. Understanding `.DAT` Files in Accounting Systems

Challenges & Structure

- **Data Structure:** A `.DAT` file can be text-based (delimited by commas, tabs, or pipes) or fixed-width binary.
- **Challenges:** Missing headers, mixed data formats, and non-standard line breaks require initial structural parsing.

2. Importing & Parsing `.DAT` Files in Power BI Desktop



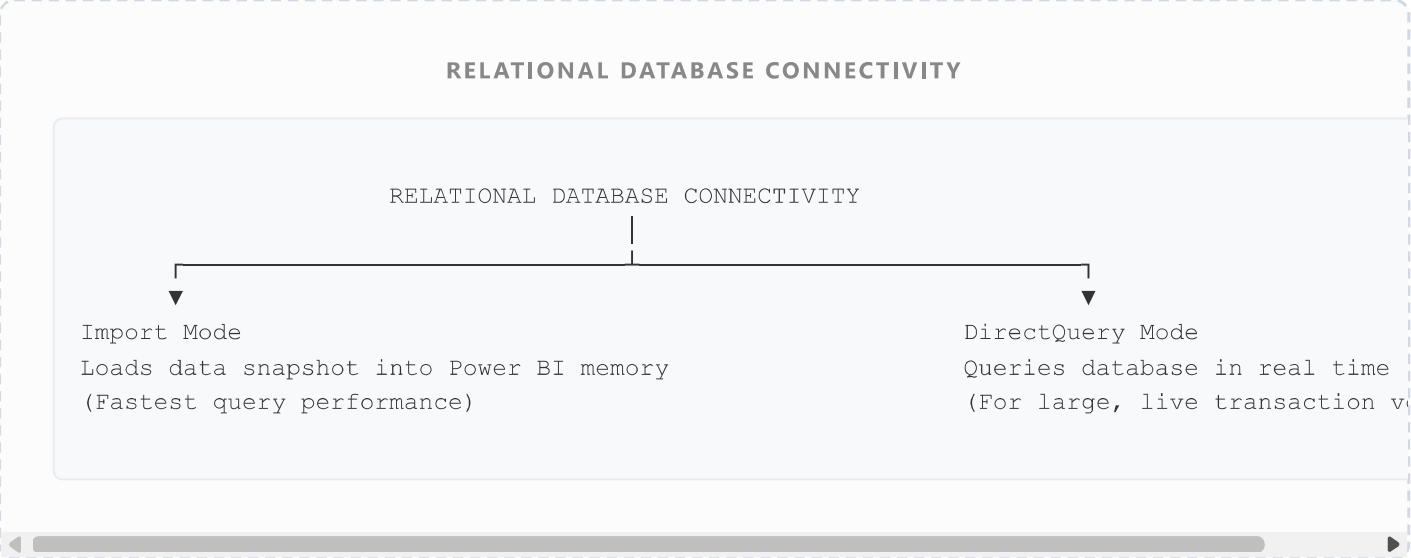
Procedure:

1. Launch **Power BI Desktop** → Click **Get Data** → Select **Text/CSV**.
2. Browse and select the legacy `.DAT` file (e.g., `PAYROLL_2026.DAT`).
3. **Delimiter Parsing:** In the preview window, specify the file delimiter (e.g., *Comma*, *Tab*, or *Custom character* like `|`).
4. Click **Transform Data** to open Power Query Editor:

- **Promote Headers:** Click *Use First Row as Headers* if the first row contains column names.
- **Set Data Types:** Change raw text strings into financial types (e.g., `Fixed Decimal Number` for Ledger Values, `Date` for Posting Dates).
- **Remove Unneeded Metadata:** Filter out trailer rows, blank balances, or system timestamps generated by legacy software.

Report Design with Relational Database Tables

Corporate accounting data frequently resides in centralized relational databases (RDBMS) like **Microsoft SQL Server**, **MySQL**, or **Oracle**. Connecting directly to database tables eliminates manual CSV exports and supports live reporting on financial ledgers.



1. Connecting Power BI to a Database

1. Go to **Home Tab** → Click **Get Data** → Choose **SQL Server Database** (or relevant RDBMS).
2. Enter the **Server Name** and **Database Name** (e.g., `FIN_DB_2026`).
3. Select the **Data Connectivity Mode**:

Leaves data in the underlying database and issues SQL queries live on demand. Useful for real-time bank ledger monitoring.

2. Designing Reports from Relational Tables

In the Navigator window, select related operational tables:

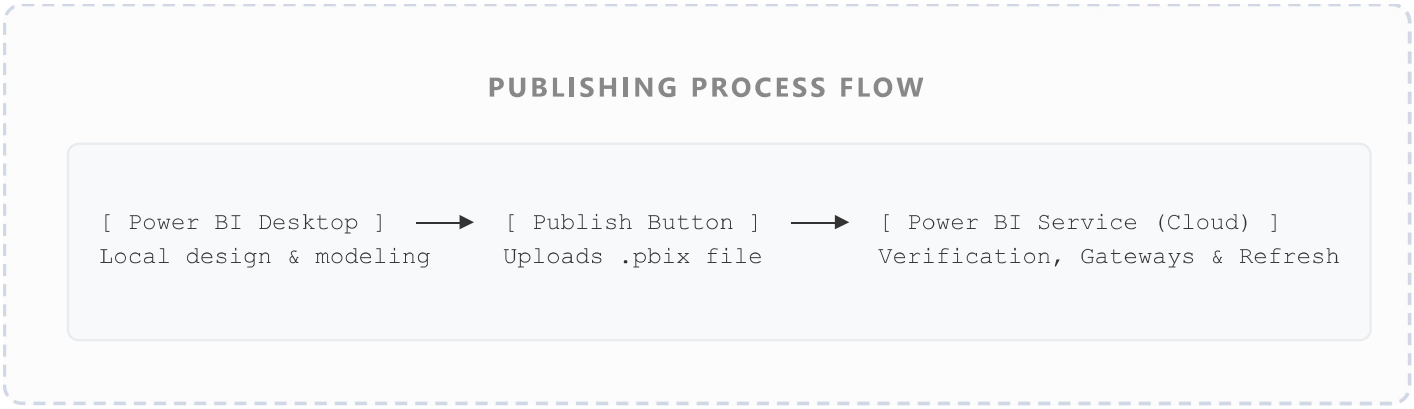
Table Types

- **Fact Table:** `Transaction_Ledger` (contains quantitative financial figures: Debit, Credit, Amount).
- **Dimension Tables:** `Chart_Of_Accounts`, `Branch_Master`, `Fiscal_Calendar` (contain descriptive lookup fields).

Power BI automatically links primary and foreign key relationships (e.g., `Account_ID` in `Transaction_Ledger` mapped to `Account_ID` in `Chart_Of_Accounts`), allowing you to drag and drop fields straight into report visuals like Matrices, Slicers, and KPI Cards.

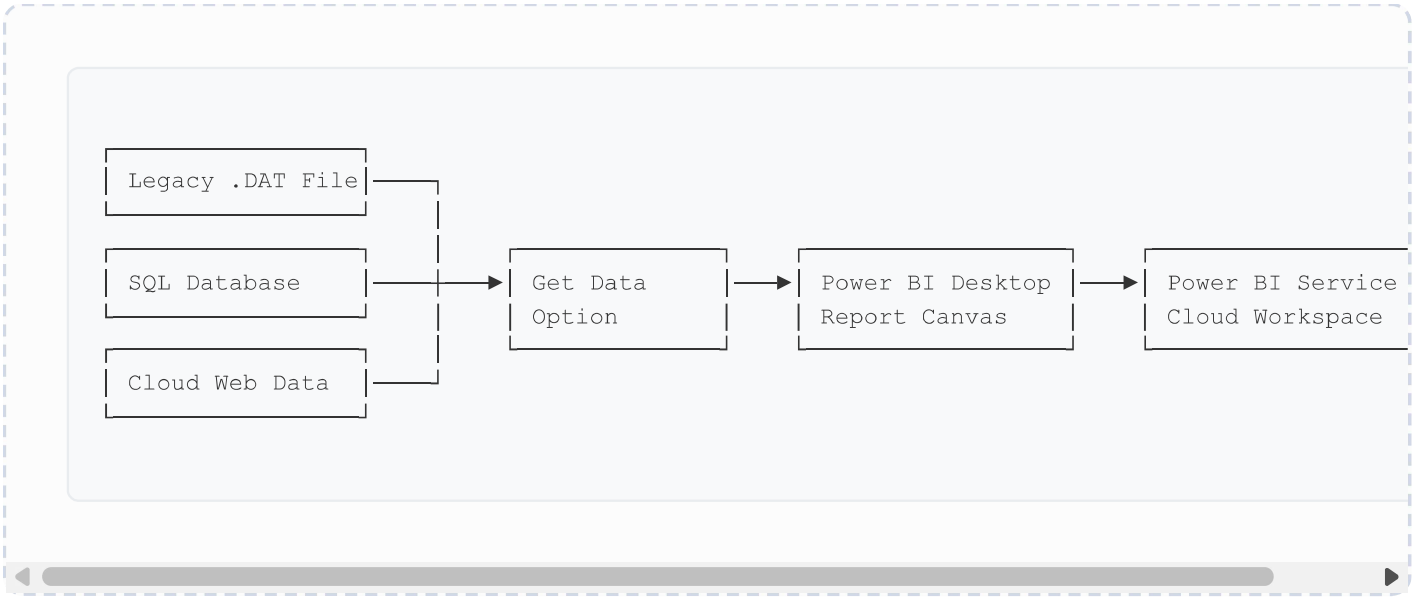
"Get Data" Processing & Cloud Verification

Designing a report on Power BI Desktop is the local phase of report development. To make financial reports accessible to senior management, statutory auditors, and board members, reports must be published and verified in the **Power BI Service (Cloud)**.



1. The Unified "Get Data" Pipeline

The **Get Data** option acts as the universal gateway for pulling data from diverse sources—whether flat legacy `.DAT` files, Excel workbooks, or relational database tables—into a standardized report model.



2. Publishing to the Power BI Cloud Workspace

1. Layout → **2. Publish** → **3. Sign In** → **4. Upload**

1. Complete the visual layout in **Power BI Desktop** (e.g., create an *Executive Financial Summary* dashboard).
2. Click **Publish** on the **Home Tab**.

3. Sign in using organizational credentials and select the target workspace (e.g., `Finance_Audit_Workspace`).
4. Power BI uploads the report layout and dataset (`.pbix` file format) to the cloud tenant.

3. Report Verification & Cloud Validation Audit

Once published, financial analysts must perform **Cloud Verification** to ensure data accuracy, proper authorization, and automated refresh schedules:

Compare total ledger balances displayed on the Cloud portal against source database query outputs to verify that data wasn't truncated during upload.

If the report relies on local `.DAT` files or local SQL Server instances, install and configure an **On-Premises Data Gateway** so the cloud service can reach local network data securely.

Set up automated refreshes (e.g., daily at 6:00 AM) under **Dataset Settings** in the cloud portal, and inspect the **Refresh History** log to ensure zero pipeline failures.

Test security roles in the cloud (e.g., verifying that the Mumbai Branch Manager can only see Mumbai branch ledger entries).

Summary Matrix of Report Design & Data Source Concepts

Data Source / Process	Connection Technique	Primary Transformation Requirement	Primary Audit & Verification Concern
Legacy `.DAT` Files	`Get Data → Text/CSV`	Custom delimiter parsing & row-level header promotion	Validating encoding, string lengths, and removing trailer rows
Relational DB Tables	`Get Data → SQL Server` (Import / DirectQuery)	Setting key relationships between Fact and Dimension tables	Verifying query folding performance and live database connection logs
Cloud Publishing	Desktop `Publish` button	Formatting report pages for web and mobile browsers	Checking On-Premises Gateway status and scheduled refresh history
Cloud Verification	Power BI Service Portal	Workspace access rights & Row-Level Security (RLS) assignments	Reconciling cloud dashboard KPI totals against source ERP balances

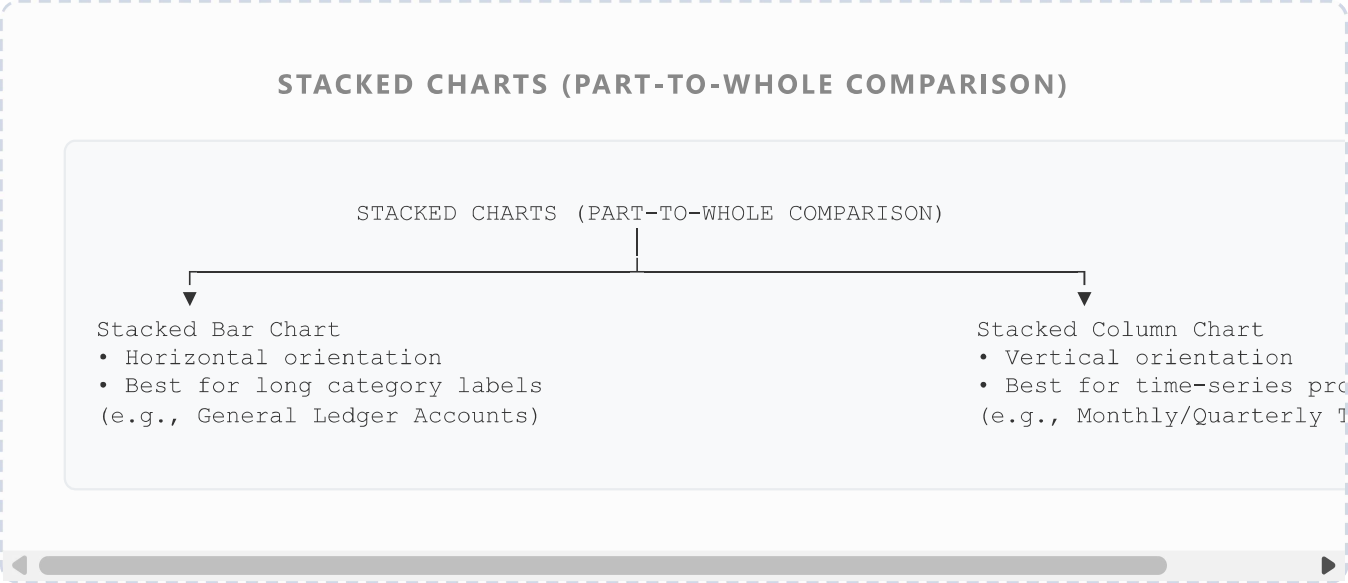
Visual Analytics & Report Design

Charting Techniques and Power BI Desktop — S.Y. B.Com (Accounts & Finance)

Unit 1: Stacked Charts | Unit 2: Clustered Charts | Unit 3: Titles & Formatting | Practical Exercise

Unit 1: Stacked Bar Charts vs. Stacked Column Charts

Stacked charts display proportional contributions of sub-categories relative to an overall total across discrete groups.



1. Stacked Bar Chart (Horizontal Layout)

Visual Structure

Categories are listed vertically along the Y-axis, while total numeric values extend horizontally along the X-axis. Each bar is segmented into color-coded sub-categories.

Best Use Case

When category names are lengthy (e.g., long general ledger account titles) to avoid text truncation.

Analyzing total operating expenses across different departments (`HR`, `Finance`, `Operations`), where each horizontal department bar is segmented into cost components (`Salaries`, `Rent`, `Utilities`, `Travel`).

2. Stacked Column Chart (Vertical Layout)

Visual Structure

Categories are listed horizontally along the X-axis, while numeric values extend vertically along the Y-axis.



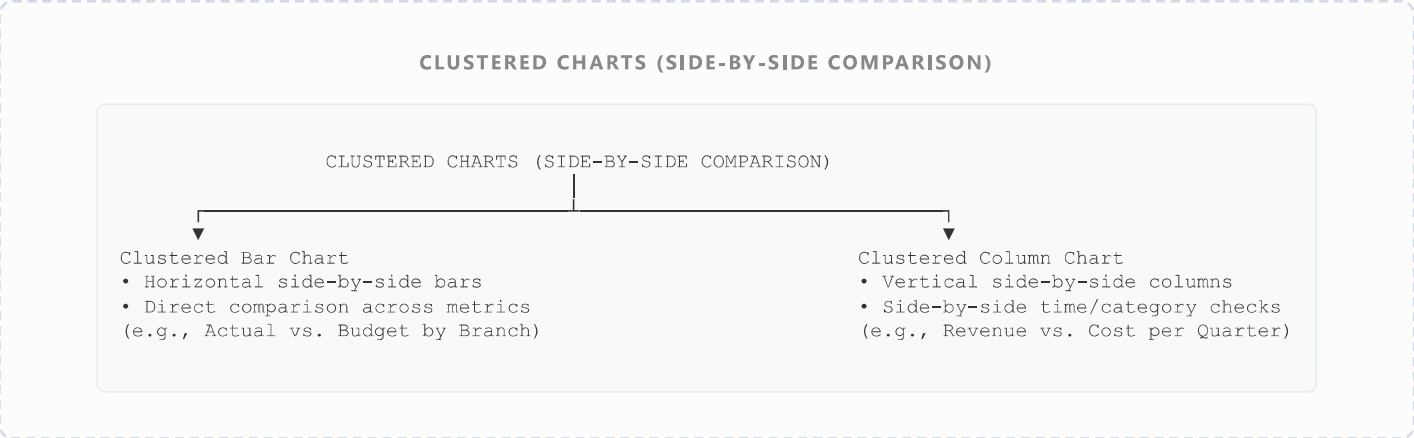
Best Use Case

Visualizing part-to-whole relationships over chronological periods (Time-Series Data).

Tracking quarterly revenue across `Q1`, `Q2`, `Q3`, and `Q4`, where each vertical column is stacked by product lines (`Software Licenses`, `Consulting Services`, `Maintenance Contracts`).

Clustered Bar Charts vs. Clustered Column Charts

Unlike stacked charts (which combine segments into a single bar), clustered charts place sub-categories side-by-side. This layout makes it easier to compare sub-categories directly against one another.



1. Clustered Bar Chart (Horizontal Layout)

Visual Structure

Displays sub-categories as parallel horizontal bars grouped under each main category heading.

Comparing `Actual Expenses` vs. `Budgeted Expenses` across five regional branch offices. Side-by-side horizontal bars quickly highlight branches that exceeded their budget limits.

2. Clustered Column Chart (Vertical Layout)

Visual Structure

Displays sub-categories as parallel vertical columns grouped side-by-side across the horizontal axis.

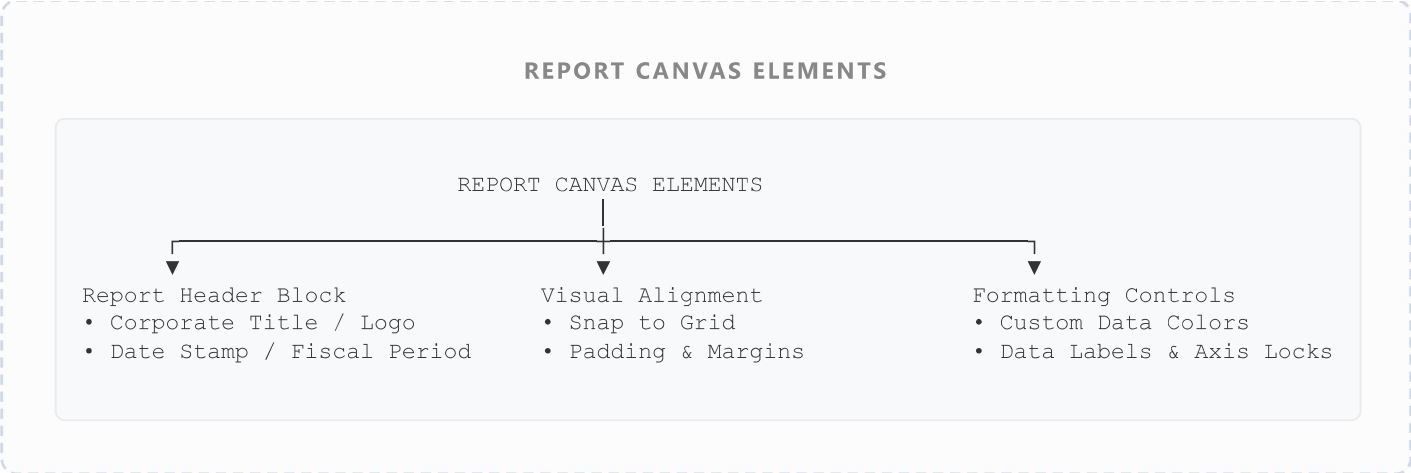
Comparing `Gross Revenue` vs. `Net Revenue` side-by-side across three consecutive financial years (`FY24`, `FY25`, `FY26`).

Comparative Summary: Stacked vs. Clustered Charts

CHART TYPE	ORIENTATION	SUB-CATEGORY PLACEMENT	PRIMARY ANALYTICAL PURPOSE
Stacked Bar	Horizontal	Stacked end-to-end within one bar	Part-to-whole breakdown for long category labels
Stacked Column	Vertical	Stacked end-to-end within one column	Part-to-whole breakdown over time periods
Clustered Bar	Horizontal	Placed side-by-side	Direct value comparison across categories
Clustered Column	Vertical	Placed side-by-side	Direct period-over-period metric comparisons

Adding Report Titles, Header Elements, and Canvas Formatting

A raw chart layout requires clear titles, defined boundaries, and standardized alignment to meet executive and statutory presentation standards.



1. Creating Professional Report Titles & Header Cards



1. Navigate to the **Insert Tab** in Power BI Desktop.
2. Click **Text Box** and place it at the top of the report canvas.
3. Type a descriptive corporate header (e.g., `ABC CORPORATION — EXECUTIVE FINANCIAL DASHBOARD (FY 2025-26)`).
4. **Formatting Standards:** Set font to a clean Sans-Serif type (e.g., *Segoe UI*), bold key labels, use `18pt–20pt` size for readability, and align text left or centered.

2. Canvas Formatting & Visual Styling Options

To format any chart, select the visual and navigate to the **Format Visual Pane** (paintbrush icon):

- **Title:** Enable visual titles and rename default auto-generated titles (e.g., change *Sum of Revenue by Branch and Quarter* to *Quarterly Branch Revenue Performance*).
- **Effects & Background:** Apply a solid white or light-gray fill, enable subtle **Drop Shadows**, and turn on **Visual Borders** with rounded corners (4px–6px) to frame individual chart elements cleanly.

- **Data Labels:** Turn on data values directly above columns or inside bars so users can read figures without hovering.
- **X-Axis / Y-Axis Controls:** Fix axis start/end values, set display units to Thousands (`K`) or Millions (`M`), or toggle gridlines.
- **Data Colors:** Apply consistent color schemes across the report (e.g., Dark Blue for `Actuals`, Light Gray for `Budget`, Red for `Unfavorable Variances`).

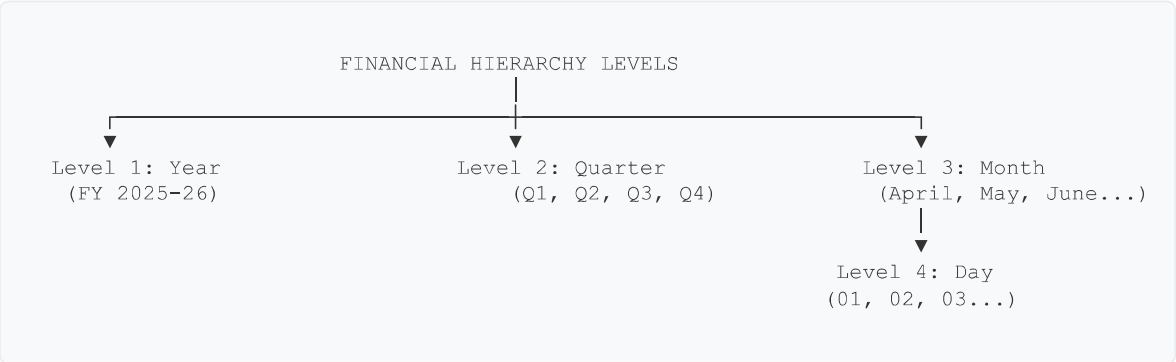
Hierarchies & Drill-Down Analysis

Dynamic Parameters and Data Filtering in Power BI Desktop — S.Y. B.Com (Accounts & Finance)

Unit 1: Hierarchies and Drill-Down Options

A **Hierarchy** in Power BI is a nested arrangement of related categorical fields organized into parent-child levels. Hierarchies allow financial analysts and auditors to start at a high-level summary and progressively explore deeper transactional layers without cluttering the report canvas.

FINANCIAL HIERARCHY LEVELS



1. Types of Accounting Hierarchies

- **Date Hierarchy:** `Year` → `Quarter` → `Month` → `Day`
- **Organizational Hierarchy:** `Division` → `Region` → `Branch Office` → `Cost Center`
- **Financial Account Hierarchy:** `Financial Statement` → `Main Group (Assets/Liabilities)` → `Sub-Group (Current Assets)` → `Ledger Account (Petty Cash)`

2. Creating a Custom Hierarchy

1. In the **Fields / Data Pane**, locate the top parent field (e.g., `Financial Category`).
2. Right-click the field → Select **Create Hierarchy**.
3. Drag child fields (e.g., `Sub-Category`, `Account Name`) into the newly formed hierarchy group, or right-click child fields and select **Add to Hierarchy**.

3. Navigation Controls for Drill-Down Analysis

When a hierarchy is added to a chart axis (e.g., a Clustered Column Chart), four interactive drill buttons appear in the top/bottom header bar of the visual:

DRILL CONTROLS



Navigates one level higher in the hierarchy (e.g., from `Month` view back up to `Quarter` view).

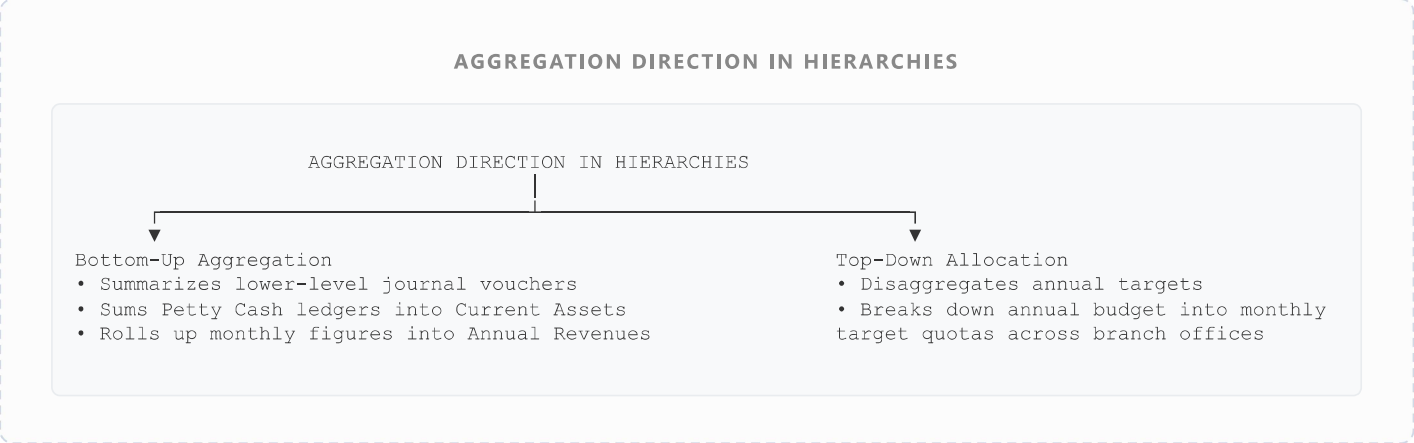
Enables selection mode. Clicking a specific data bar (e.g., `FY26`) filters the visual to show the quarters of *only* that selected year.

Steps down one level across all categories simultaneously (e.g., moves from displaying `Years` to displaying combined `Quarters` for all years).

Retains the parent context while adding the child level underneath (e.g., displaying `2026 Q1`, `2026 Q2`, `2026 Q3`).

Aggregation Rules with Bottom-Up & Top-Down Navigation

When users move up or down through hierarchy levels, Power BI recalculates numeric measures on the fly using **Aggregation Rules**.



1. Bottom-Up Aggregation (Summarization)

Bottom-up navigation starts at the granular transaction level and rolls numeric values up into higher categories using mathematical functions (`SUM`, `AVERAGE`, `COUNT`).

> **Accounting Example**
> Individual journal entries for `Office Stationery`, `Printer Ink`, and `Tea Expenses` roll up into the `Administrative Expenses` ledger, which rolls up into `Total Operating Expenses (OPEX)` on the Income Statement.

2. Top-Down Navigation Rules (Disaggregation & Context)

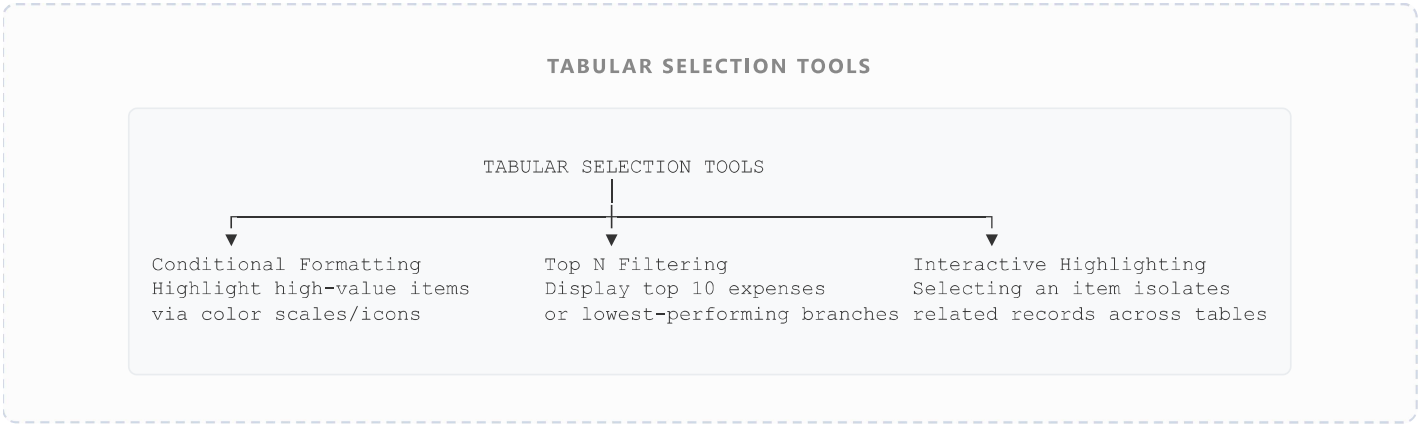
Top-down navigation breaks high-level totals down into underlying components. When drilling down, the context of the parent category filters all child items automatically.

Rule of Additivity

Balance sheet items like `Cash` and `Inventory` are **additive across categories** (summed across branches), but **non-additive across time** (closing cash balance for December is the end-of-month figure, not the sum of daily balances).

Automated Records Selection with Tabular Data

In financial auditing, manually searching through thousands of journal entries is inefficient. Power BI provides automated selection tools within **Table** and **Matrix** visuals to highlight and isolate records based on rules.



1. Conditional Formatting for Audit Exceptions

Auditors use conditional formatting to highlight unusual ledger transactions automatically:

- **Background Color / Font Color:** Applies a red fill to any expense entry that exceeds its budgeted threshold or variance limit (e.g., `Variance > ₹ 50,000`).
- **Data Bars:** Displays horizontal bars inside matrix cells to visualize relative account balances.
- **Icon Sets:** Displays alert icons (e.g., Red Warning Triangles for overdue invoices past 90 days, Green Checkmarks for settled accounts).

2. Top N Selection Rules

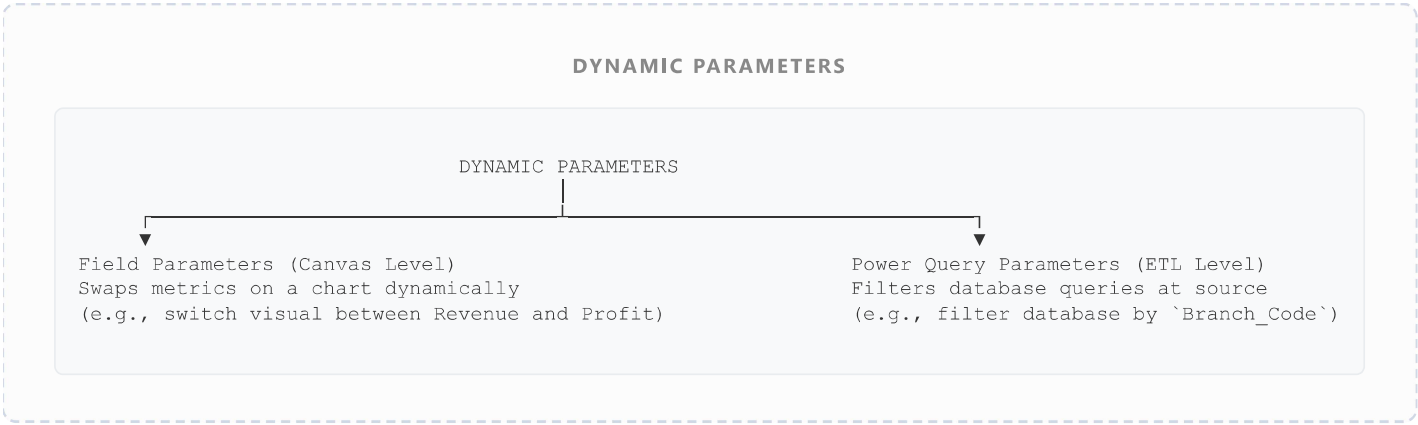
Filter the visual to display only the top or bottom records automatically without altering the underlying dataset:



1. Open the **Filters Pane** → Expand a categorical field (e.g., `Vendor Name`).
2. Set **Filter Type** to **Top N**.
3. Set **Show items:** `Top 10` → Drag `Expense Amount` into the **By Value** field → Click **Apply Filter**.
4. *Result:* Automatically isolates the top 10 vendor payables for expenditure review.

Parameters for Field Data, Tables, and Query Filters

Parameters allow report users to change report fields, switch calculation models, or filter database queries dynamically without editing underlying code.



1. Field Parameters (Dynamic Axis & Measure Selection)

Field parameters allow users to switch which dimension or measure is displayed on a chart using a simple selection slicer.

Accounting Application

A single chart can switch between displaying `Sales Revenue`, `Net Profit`, or `Operating Costs` based on a slicer choice, avoiding the need for three separate charts.

1. Go to **Modeling Tab** → Click **New Parameter** → Select **Fields**.
2. Name the parameter (e.g., `Select Financial Metric`).
3. Select fields: `[Total Revenue]`, `[Net Profit]`, `[Total Expenses]`.
4. Place the generated parameter field on the chart's Y-axis and display the parameter slicer on the canvas.

2. Query Parameters in Power Query (Data Source Filtering)

Query parameters are defined in the Power Query Editor to filter incoming database tables before data is loaded into memory.

Use Case

Restricting a database query to import data for a specific financial year or branch code (e.g., `Parameter_Region = 'Mumbai'`).

1. Open **Power Query Editor** → Click **Manage Parameters** (Home Tab) → **New Parameter**.
2. Define Name (`BranchParam`), Type (`Text`), and Current Value (`MUMBAI`).
3. Filter the `Branch` column in the data query using `BranchParam`.
4. Changing the parameter value updates the entire dataset filter automatically during data refresh.

Summary Matrix of Advanced Power BI Report Features

FEATURE / CONCEPT	MENU LOCATION	PRIMARY FUNCTION	ACCOUNTING & AUDIT VALUE
Custom Hierarchy	Fields Pane → Right-click Field	Nests fields into parent-child levels	Drills down from Annual summaries to Voucher details
Drill Controls	Visual Header Bar	Navigates up/down through visual levels	Explores regional and departmental cost centers
Conditional Formatting	Format Pane → Cell Elements	Applies dynamic colors, bars, and icons	Highlights budget overruns and audit exceptions
Top N Filter	Filters Pane → Filter Type	Restricts view to top or bottom entries	Identifies top 10 accounts receivable or expense items
Field Parameters	Modeling Tab → New Parameter	Swaps chart axes or metrics dynamically	Toggles visuals between Revenue, Expense, and Profit
Query Parameters	Power Query → Manage Parameters	Filters source database queries	Ingests data for a single branch or fiscal year